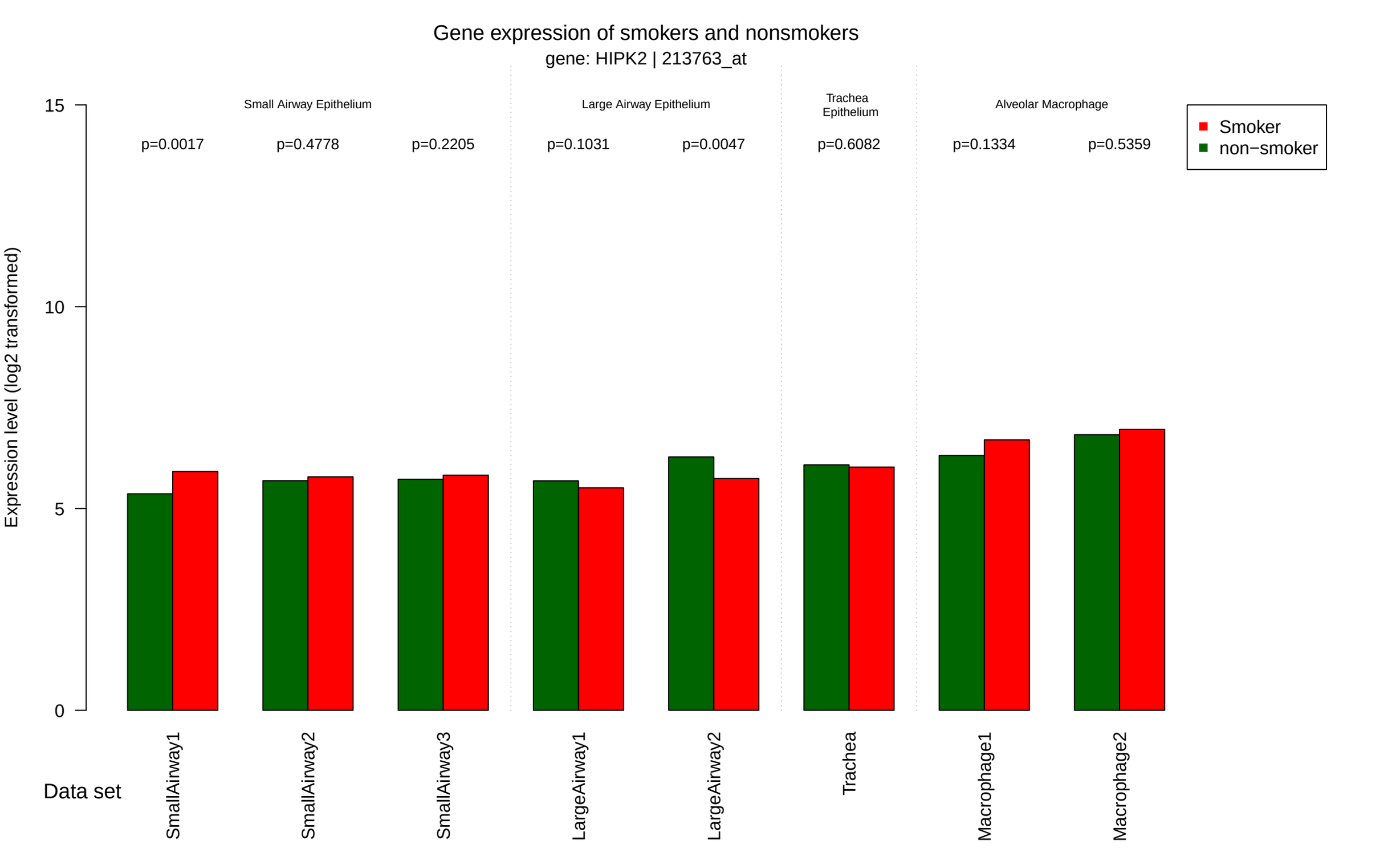
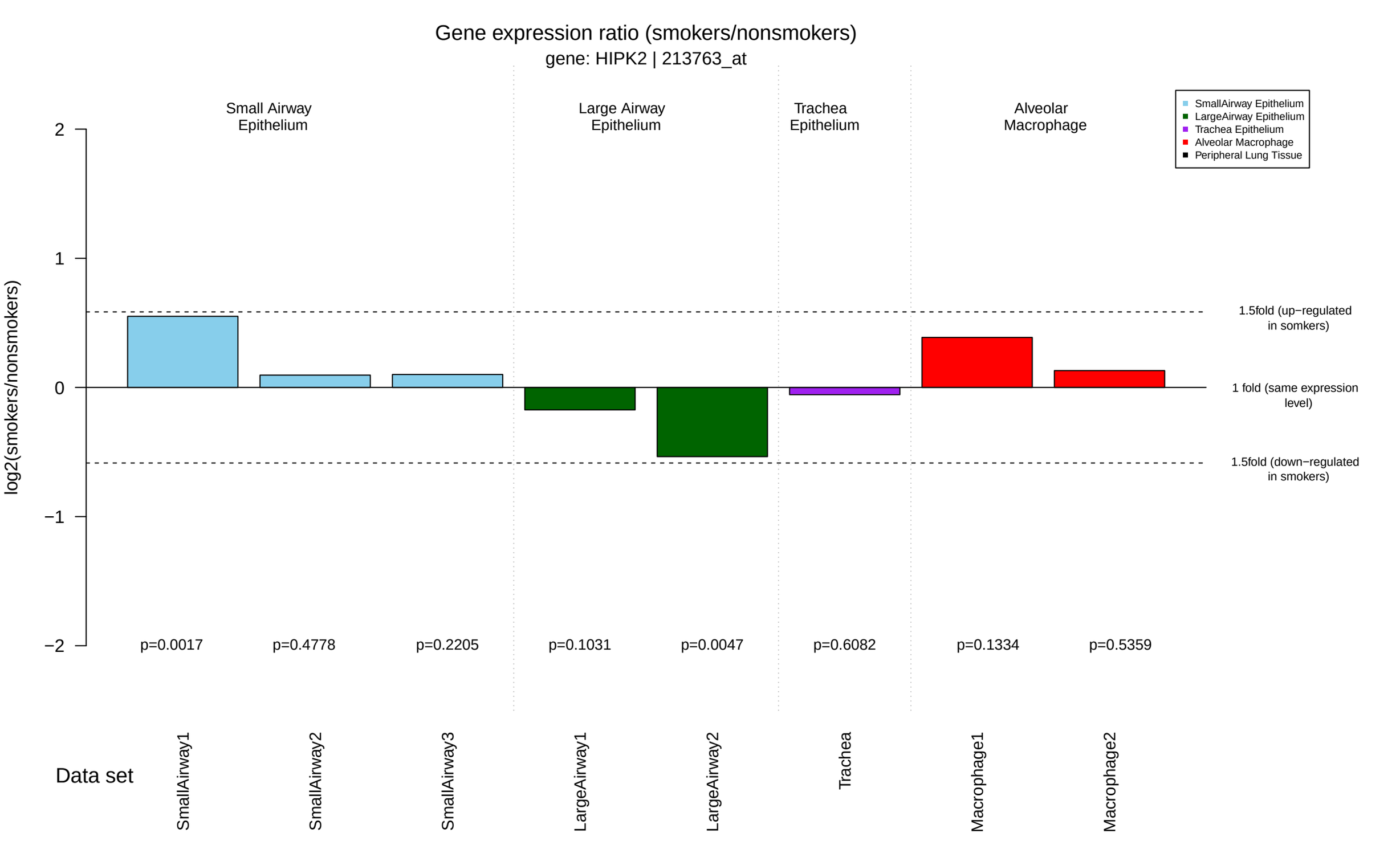
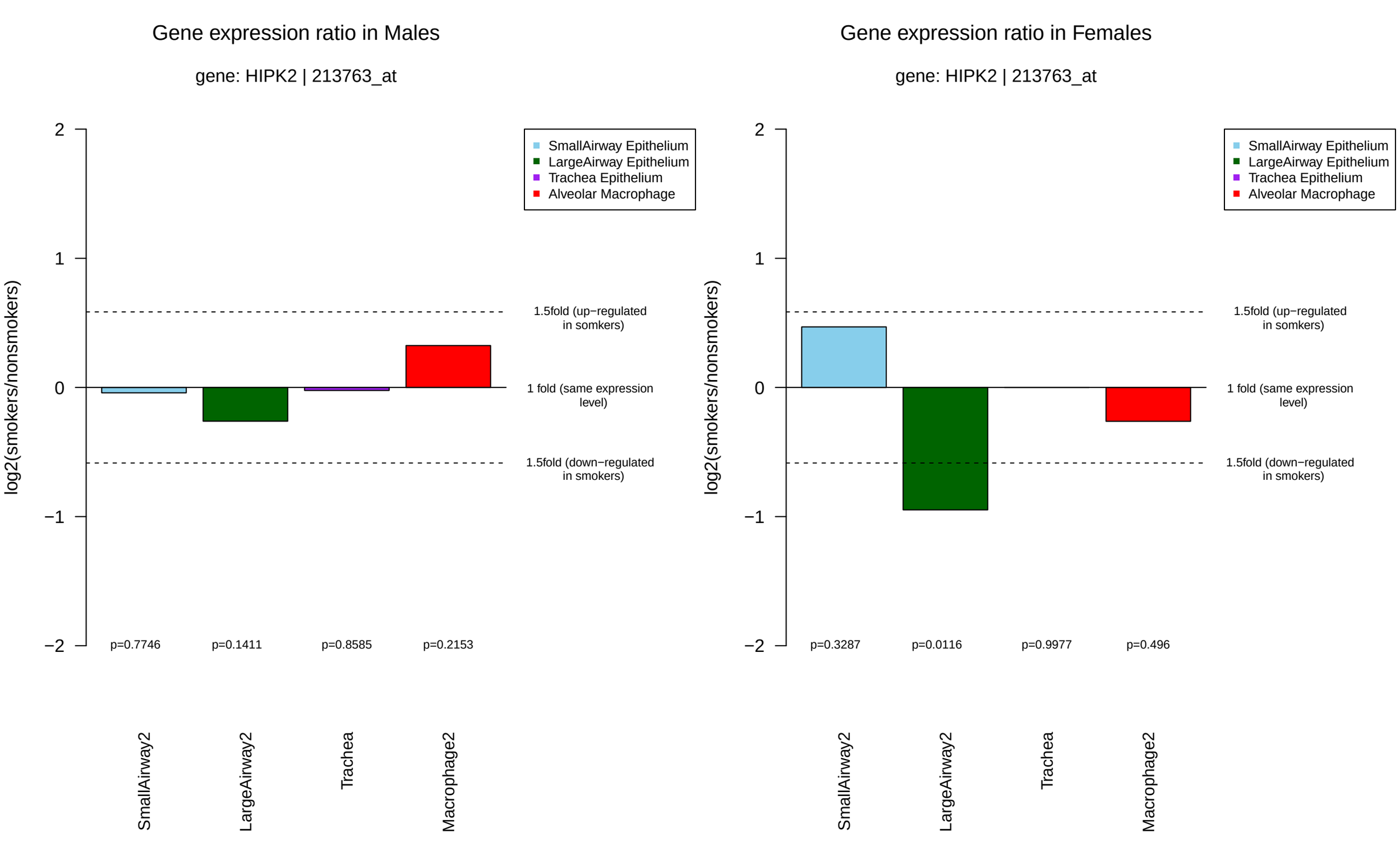
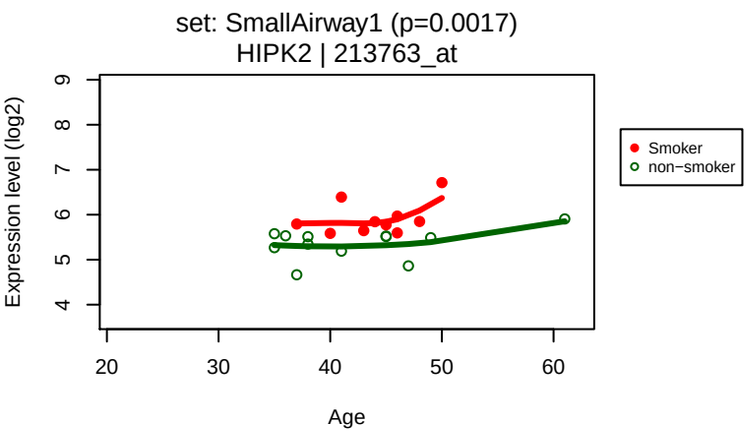
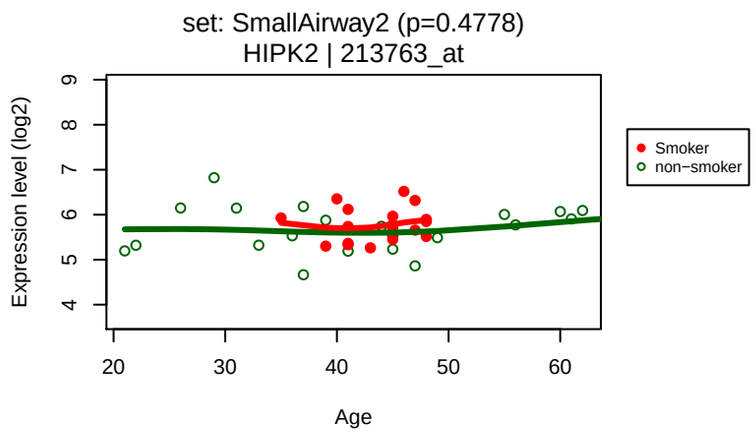
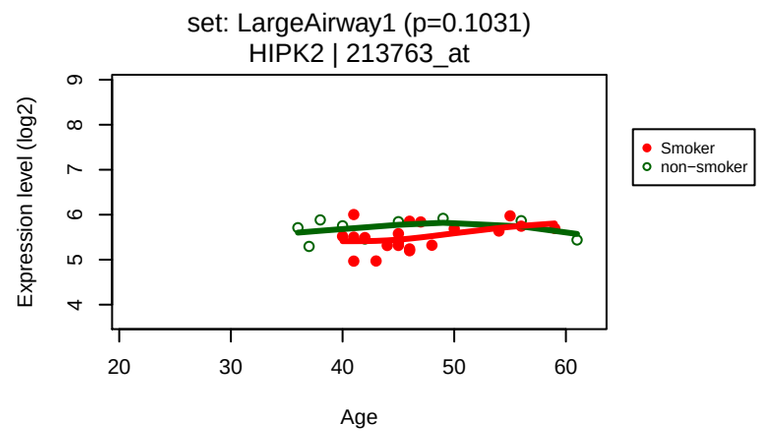

Gene expression of smokers and nonsmokers
gene: HIPK2 | 213763_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0017
p=0.4778
p=0.2205
p=0.1031
p=0.0047
p=0.6082
p=0.1334
p=0.5359
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

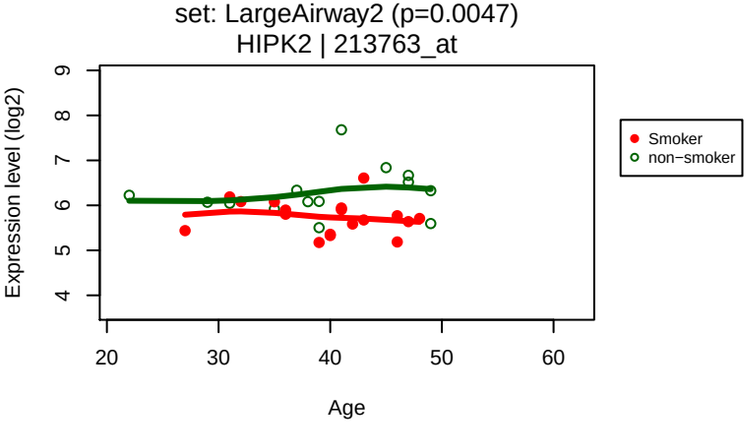

Gene expression ratio (smokers/nonsmokers)
gene: HIPK2 | 213763_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.4778
p=0.2205
p=0.1031
p=0.0047
p=0.6082
p=0.1334
p=0.5359
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

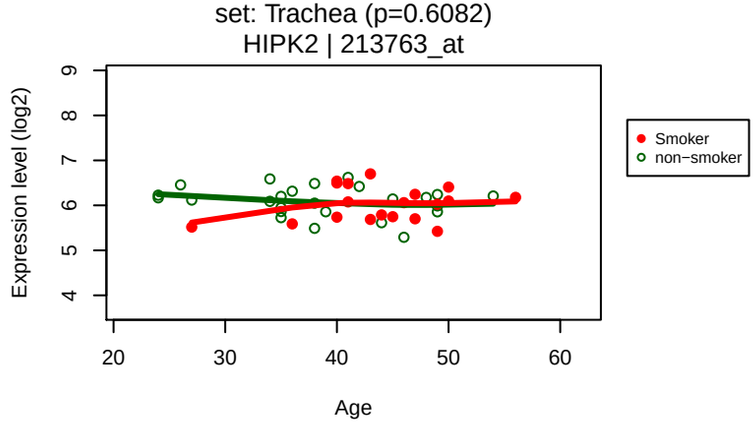

Gene expression ratio in Males
gene: HIPK2 | 213763_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7746
p=0.1411
p=0.8585
p=0.2153
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK2 | 213763_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3287
p=0.0116
p=0.9977
p=0.496
SmallAirway2
LargeAirway2
Trachea
Macrophage2

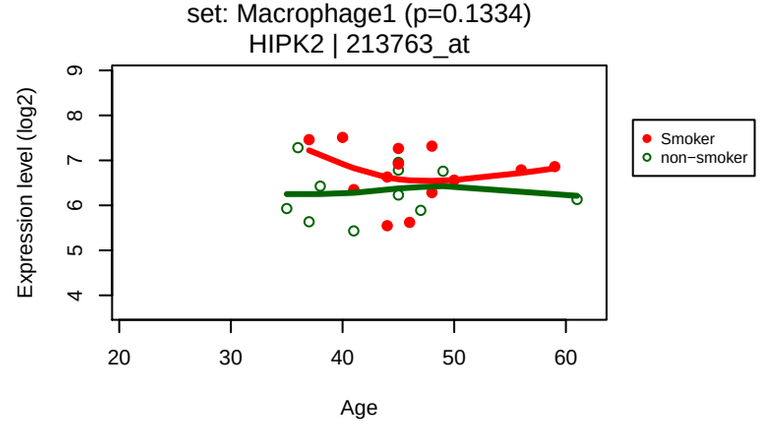

set: SmallAirway1 (p=0.0017)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker

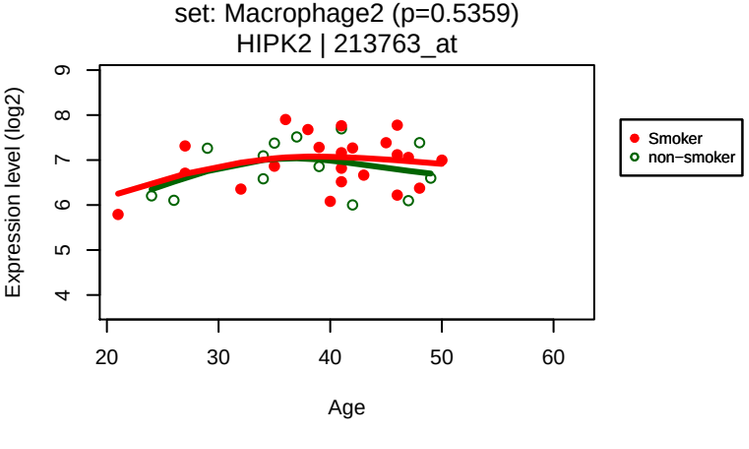

set: SmallAirway2 (p=0.4778)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker

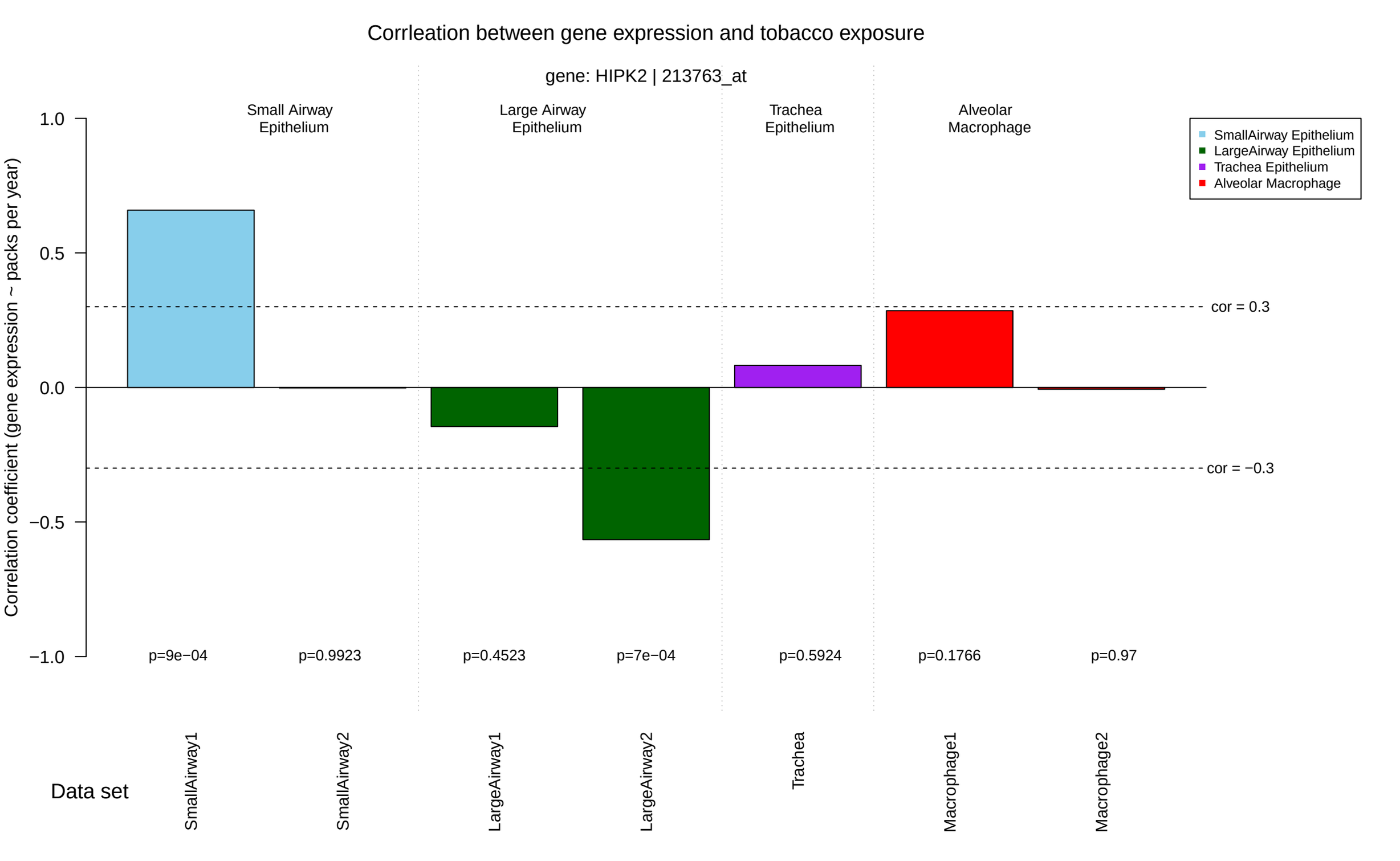

set: LargeAirway1 (p=0.1031)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0047)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6082)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1334)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5359)
HIPK2 | 213763_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK2 | 213763_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=9e−04
p=0.9923
p=0.4523
p=7e−04
p=0.5924
p=0.1766
p=0.97
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2